MyoDishAnalysis quick start: the watcher

Version 1.0.0-beta.5 · automatic analysis of all recordings of a folder · full description: README, *help MyoDishAnalysisWatch*, *mda-watch --help*

What it does

Every pass searches the raw folder and its subfolders for *.mdd* recordings and analyses the new or changed ones: every contraction, a summary per time bin (60 min) outside the stimulation protocols, and the protocols separately. The raw folder is only read. MATLAB and Python write the same results and share the index.

Start

```
One pass:  MyoDishAnalysisWatch('/data/myodish/raw', '/data/myodish/results')
           mda-watch /data/myodish/raw /data/myodish/results                (Python)
Schedule:  MyoDishAnalysisWatch(RAW, RESULTS, 'schedule', 'daily 10:00')   ('status' / 'stop')
           mda-watch RAW RESULTS --schedule "daily 10:00"     (Python, until Ctrl+C)
           'weekly Sun 05:00'  'Mon,Thu 06:30'  'every 2 days 10:00'  'every 120 min'  'every 6 h'
Options:   'rockerFilter', true   'fromDate', '2026-10-01'   'dryRun', true   'resultFormat', 'csv'
           'include', 'ABC00*' (folder / file names, * ? wildcards)   'exclude', '*test*'
           Python: --include "ABC00*" --workers 6 (parallel processes)
```

Without an open MATLAB: the scheduler of the operating system with *mda-watch RAW RESULTS --quiet*. Checked again in the next pass: changed in the last 10 min, *running* (no *Recording stopped*, data end < 30 h ago), *incomplete* (file shorter than in the log file; analysed when unchanged for 30 h).

Results folder (one folder per experiment, setup and recording)

file	content
ABC000101/ABC000101_overview.png	**start here**: all setups × channels: amplitude since the start, status of the slice (colour), last 24 h, red flags (red frame, also of earlier days)
ABC000101/ABC000101_report.pdf	analysis report of all recordings of the experiment (see below)
ABC000101/Setup1_ABC000101/ Setup1_ABC000101_ch<c>.png	**course per channel**: since the start (slice changes, red flags) and the last 24 h (comments); data: …_trends.xlsx
ABC000101/Setup1_ABC000101/02/	results of recording Setup1_ABC000101_2: **<name>.xlsx** (summary per channel and hour, trends, channels, gaps, events, info), _contractions / _pulses / _overview.csv.gz, _protocols.xlsx (FFR, PRP, thresholds), trend images <name>_ch<c>.png
ABC000101_slices.csv, mda_slices.csv	slice register: one row per slice from putting it in until it was taken out
mda_index.csv, mda_flags.csv	one row per recording (status, version, message); flags per recording and channel (GUI: **Results …**)

The analysis report

Page 1: the overview image. Then all recordings of the experiment × channels with capture (%) and flag codes; the recordings of the last pass in **bold**, earlier ones keep their flags. **Red** = check the analysis (it may not work, or technical faults put wrong values into the data): **I** inverted signal (sensor fault), **N** noise suspected (uncertain ≥ 25 % with extra + missed beats ≥ 15 %), **S** large extra beats (spikes), **D** delay pulse → onset > 150 ms, **F** pacing > 10 Hz (status channel misread). **Yellow** = slice quality (biology; you decide): **C** capture < 90 %, **X** extra beats > 10 %, **A** amplitude < 75 µN, **T** amplitude changes by > 30 % during the recording, **V** amplitude CV > 30 %. Grey: fewer than 50 contractions. Arrhythmic pacing: capture with ~, no C and V. Then the list with values and channel labels and traces of the red channels of the last pass with the command that opens exactly this place in the GUI:

```
MyoDishAnalysisGUI('<results>/<name>.xlsx', 'channel', 3, 'window', [1200 1220])
mda-gui "<results>/<name>.xlsx" --channel 3 --window 1200 1220
```

MATLAB also prints these commands as links in the command window. Flags and thresholds: options *flagExtraBeats*, *flagUncertain*, *flagCapture*, … (Python: *flag_…*); *'report'*, *false* / *--no-report*: no PDF.

Good to know

- **After an update** the earlier results are kept and only new and changed recordings are analysed (the report counts those of another version or with other options); *'reanalyze'*, *'outdated'* (*--reanalyze outdated*) analyses them again, *'all'* every recording.
- Errors stop only the recording concerned (status *error*); retried when the file or the version changes.
- **Parallel recordings** (schedule files, snapshots) are not analysed when their main recording (same folder) is analysed in the same pass (status *parallel*): their data and protocols are in the main file's results. Only the schedule files: *'include'*, *'*schedule*'* (analysed and marked; the slice register counts every beat once).
- *'folders'*, *'flat'*: all files of an experiment in one folder as before; *MyoDishAnalysisWatch('reorganize', RESULTS)* (*mda-watch RESULTS --reorganize*) moves earlier results into the folders of the setups. *'resultFormat'*, *'csv'*: one csv file per table. Images again: *MyoDishAnalysisWatch('trends', RESULTS)*.
- **Before / after a substance**: *mda_timepoints(RESULTS, 'Mava100', 'Setup1_ABC000101', [1 3], '2000-01-01 15:20', [-60 120])* (*mda-timepoints*): windows of 120 s at the given minutes, % of baseline, notes, images; files in *ABC000101/timepoints/*.
- One watcher per results folder (lock file *mda_watch.lock*). Labels per channel: *<name>_labels.csv* next to the recording (saved by the GUI). GUI settings: *'settings'*, *file* (Advanced … → Save settings).